Computer Networks : Assignment 1.

→ design a simple lan using 4 different topologies & test it using ping utility.

→ theory :

- LAN :

it is a network that connects computers & devices within a limited geographical areas. LAN enables us to share data, files & internet connection at high speed.

- network topologies :

1. bus topology :

all devices are connected to a single communication cable called a backbone cable.

pros : easy to install
low cost

cons : failure causes entire network stoppage
difficult to troubleshoot
limited cable lenght

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2. star:

every device is connected to a central hub / switch.

adv: easy to install & manage
easy fault detection
high performance

disadv: requires more cable
cost is higher
failure of hubs.

3. ring:

each device is connected to 2 neighbouring devices, forming a closed loop.

adv: equal access for every device
no data collision
better performance under heavy load

disadv: failure of 'one' device could stop comms.
difficult to troubleshoot
adding / removing devices effects the network.

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4. Mesh:

each device is connected to every other device.

adv: very reliable
very secure
multiple comm. paths.

disadv: very expensive
complex setup
large amount of cabling required

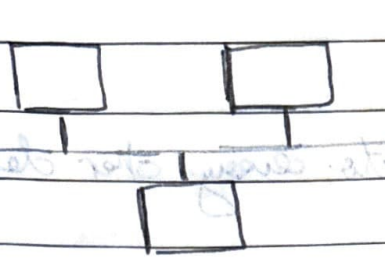
5. tree:

it combines star & bus topology in a tree hierarchical structure.

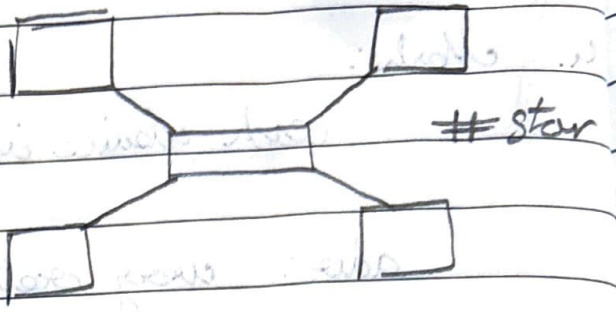
adv: easy to expand
easy to manage
suitable for large orgs

disadv: backbone failure affects many nodes.
expensive
complex wiring

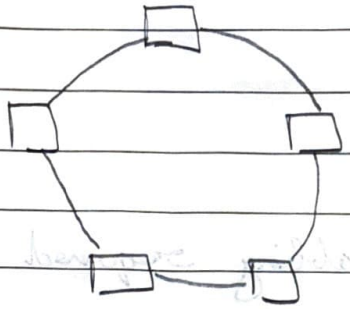
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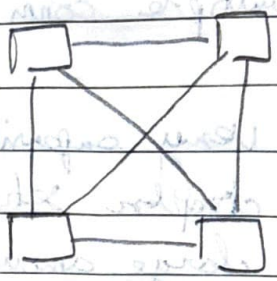
bus topology



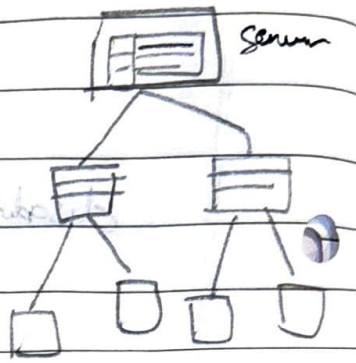
star



ring



mesh



tree

steps:

- open cisco packet tracer
- create req. topology
- add pc's & req. networking devices
- connect devices using appropriate cables
- assign IP addresses
- open desktop → cmd on a PC
- use command : ping <dest. IP>
- verify

conclusion:

The LAN was successfully created using diff. network topologies, IP addresses were assigned & verified.

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FAQ's.

1) LAN is a network that connects devices within a limited ~~area~~ geographical area such as a home, office ok. It allows users to share resources with one another. LAN's are privately owned, provide high speed communication & are easy to maintain

2. PING (packet internet groper) is a network diagnostic utility used to test connections between & devices. It sends a ICMP echo packet & waits for a ICMP echo reply from the destination.

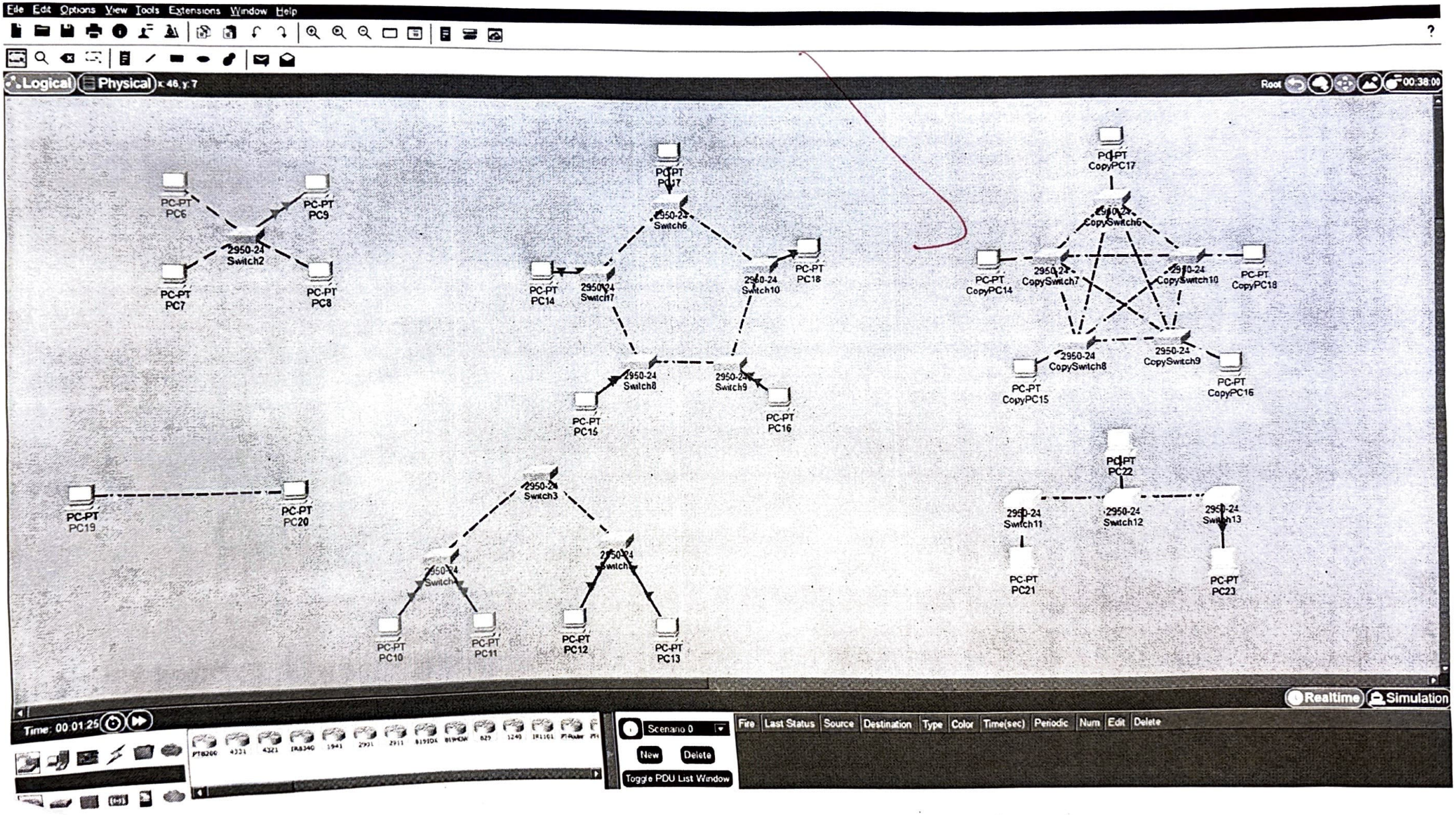
uses: check if host is reachable
verifies connectivity
measures round trip time

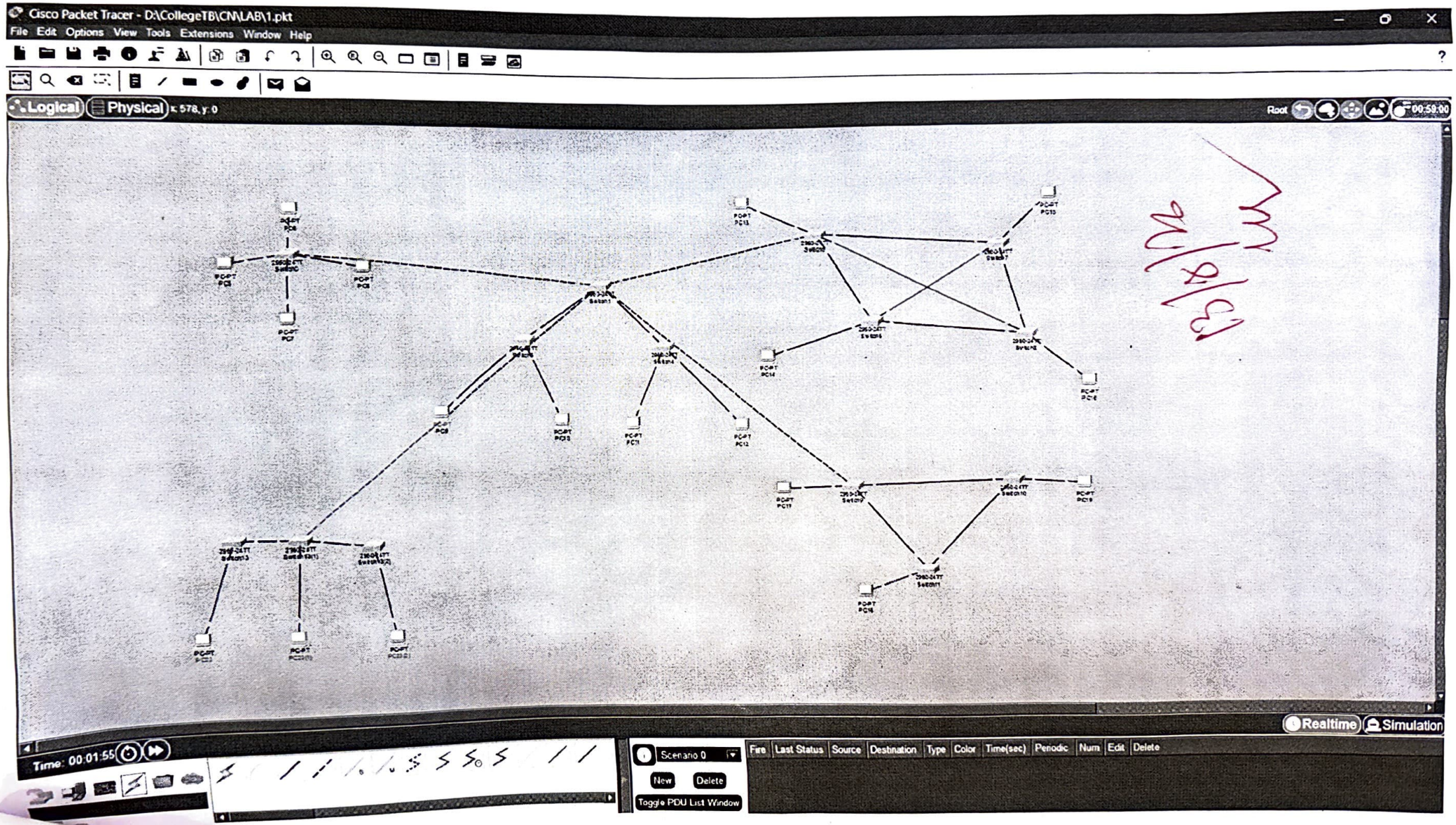
3/8/24

5.

IP Class	default subnet mask
A	255.0.0.0 (/8)
B	255.0.0.0 (/16)
C	255.255.255.0 (/24)
D	NA
E	NA

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File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 48, y: 7
Root
00:38:09
PC-PT PC6
PC-PT PC9
2950-24 Switch2
PC-PT PC7
PC-PT PC8
PC-PT PC17
2950-24 Switch6
PC-PT PC14
2950-24 Switch7
2950-24 Switch10
PC-PT PC18
2950-24 Switch8
2950-24 Switch9
PC-PT PC15
PC-PT PC16
PC-PT CopyPC17
2950-24 CopySwitch6
PC-PT CopyPC14
2950-24 CopySwitch7
2950-24 CopySwitch10
PC-PT CopyPC18
2950-24 CopySwitch8
2950-24 CopySwitch9
PC-PT CopyPC15
PC-PT CopyPC16
PC-PT PC19
PC-PT PC20
2950-24 Switch3
PC-PT PC10
PC-PT PC11
PC-PT PC12
PC-PT PC13
PC-PT PC22
2950-24 Switch11
2950-24 Switch12
2950-24 Switch13
PC-PT PC21
PC-PT PC23
Realtime
Simulation
Time: 00:01:25
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete


Cisco Packet Tracer - D:\CollegeTB\CN\LAB\1.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 578, y: 0
Root
00:59:00
Time: 00:01:55
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
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